Chaitanya Bharathi Institute of Technology (Autonomous)

Department of Computer Science and Engineering

Fundamentals of Data Science

Case Study Submission

Case Study 1 · Online Retail Shopping Baskets

Case Study 4 · Student Dropout and Academic Success

Submitted by	Neha Samanvitha Valiveti
Roll number	160124733159
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Course faculty	Smt. S. Durga Devi
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Part I — Case Study 1: Online Retail Shopping Baskets

1.1 Problem Statement and Objectives

An online retail store holds transaction records in which several products are purchased together. However, the business has no systematic analysis of transaction characteristics or of groups of similar shopping baskets. The objective is to analyse the transaction data using statistical and machine learning techniques, identify similarities between transactions, compare clustering algorithms and derive meaningful conclusions from the results.

Objectives:

- Understand the structure and characteristics of the retail dataset, and clean and preprocess the transaction records using Pandas.
- Calculate descriptive statistics, identify frequently occurring products, and visualise product frequencies and transaction-size distributions.
- Represent transactions as numerical feature vectors and apply PCA for dimensionality reduction and visualisation.
- Implement K-means, hierarchical clustering, DBSCAN and spectral clustering, and compare them using the silhouette coefficient and the Davies–Bouldin index.
- Discuss the limitations of the analysis and its business implications.

1.2 Dataset Description

The dataset (Online_retail_new.csv) contains **7,501 transactions** stored in 20 columns. Each row is one basket and each column is an item *slot*, not a fixed product: a basket with three products fills the first three slots and leaves the rest empty. The file has **no header row**. Its first line (shrimp, almonds, avocado, ...) is itself a transaction, so it was read with header = None to avoid losing that basket and mistaking product names for column names. The Excel version of the file contains the same records.

Property	Value
Transactions (rows)	7,501
Item slots (columns)	20
Non-empty item entries	29,363
Empty slots	120,657 (about 80% of the grid)
Distinct products (after cleaning)	119
Exactly repeated baskets	2,325
Baskets with an item listed twice	5

1.3 Methodology

Step 1: Cleaning and preprocessing

Empty slots were dropped, product names were stripped of surrounding spaces and converted to lower case, and items listed twice within the same basket were counted once, because a basket either contains a product

or it does not. The 2,325 exactly repeated baskets were **kept**. Popular small baskets, such as a single bottle of mineral water, are genuinely purchased many times, and deleting them would distort product frequencies.

```
baskets = raw.apply(lambda r: sorted({str(x).strip().lower()
                                      for x in r.dropna()}), axis=1)
```

Purpose: this line converts each row into a clean, de-duplicated list of product names. *Observation:* all 7,501 baskets remained non-empty, covering 119 products.

Step 2: Feature vectors

Clustering algorithms need numerical input, so each basket was one-hot encoded with MultiLabelBinarizer into a 119-length binary vector (1 = product present, 0 = absent). The resulting 7,501 × 119 matrix is **96.7% zeros**.

Step 3: Dimensionality reduction with PCA

PCA was first fitted on all components to study how variance is distributed. The data were then projected onto the first **10 principal components** for clustering. This reduces noise from rare products and makes all four algorithms practical to run.

Step 4: Clustering algorithms

- **K-means:** partitions baskets into k groups by minimising within-cluster squared distance. k was selected by comparing inertia, the silhouette coefficient and the DB index for k = 2 to 10.
- **Hierarchical (agglomerative, Ward linkage):** repeatedly merges the two clusters whose union least increases within-cluster variance, cut at the same k for a fair comparison.
- **DBSCAN:** groups points lying in dense regions and labels isolated points as noise. With min_samples = 20, eps was chosen from the k-distance graph.
- **Spectral clustering:** builds a 30-nearest-neighbour similarity graph, embeds it using the eigenvectors of the graph Laplacian and partitions it into k groups.

Step 5: Evaluation metrics

The **silhouette coefficient** (range −1 to 1; higher is better) compares each point's distance to its own cluster with its distance to the nearest other cluster. The **Davies–Bouldin index** (≥ 0; lower is better) averages, for each cluster, the ratio of within-cluster scatter to the separation from the most similar cluster. Both were computed on the same 10-dimensional PCA features. The Adjusted Rand Index (ARI) was also used to measure how much the algorithms agree with each other.

1.4 Statistical Findings

1.4.1 Transaction size

Statistic	Value
Mean items per basket	3.91
Median	3
Mode	1
Standard deviation	2.90

Statistic	Value
Minimum / Maximum	1 / 20
Skewness	1.19
Single-item baskets	23.4%

The average basket contains 3.91 items, but the median is 3 and the most frequent size is just one item. A skewness of 1.19 indicates a clear right skew: many small baskets and a few large ones (up to 20 items) that pull the mean upwards. The median is therefore a better description of a typical transaction than the mean.

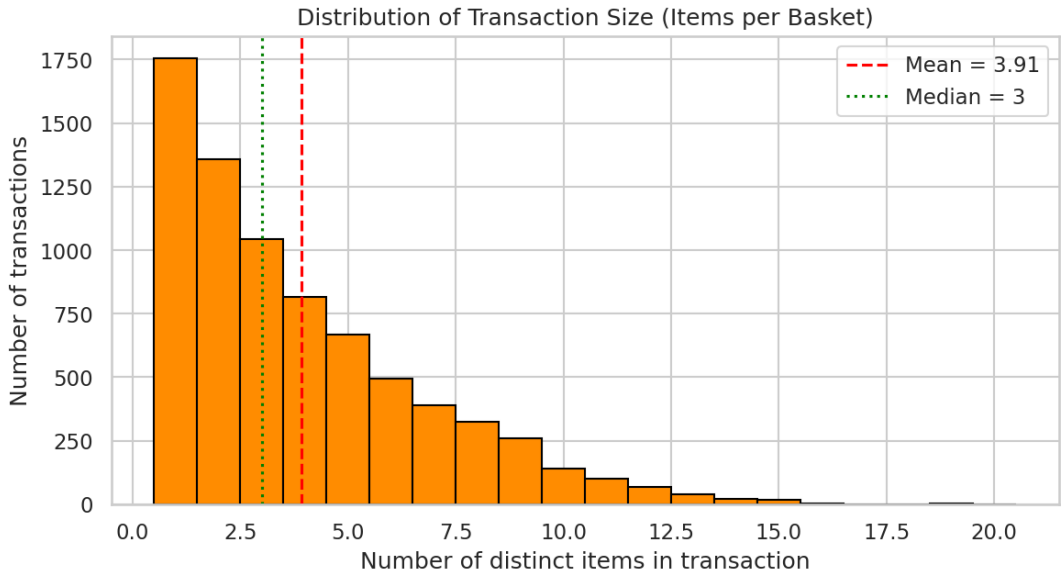


Figure 1. Distribution of transaction size (items per basket).

Interpretation: the number of transactions falls steadily as basket size increases. The mean line lies to the right of the median, confirming the right skew. Most customers make short top-up purchases rather than large weekly shops.

1.4.2 Frequently occurring products

The *support* of a product is the percentage of transactions that contain it.

Rank	Product	Transactions	Support (%)
1	Mineral water	1,788	23.84
2	Eggs	1,348	17.97
3	Spaghetti	1,306	17.41
4	French fries	1,282	17.09
5	Chocolate	1,229	16.38
6	Green tea	991	13.21
7	Milk	972	12.96
8	Ground beef	737	9.83

Rank	Product	Transactions	Support (%)
9	Frozen vegetables	715	9.53
10	Pancakes	713	9.51

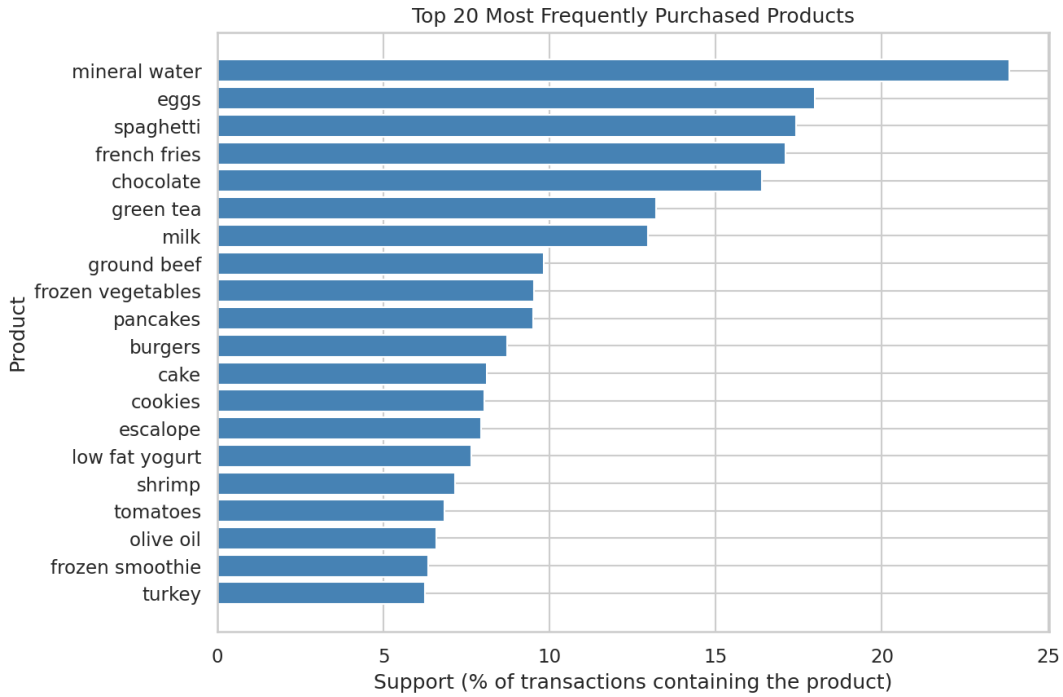


Figure 2. Top 20 most frequently purchased products by support.

Interpretation: product demand follows a long-tail pattern. Mineral water appears in almost one basket in four, and seven products each appear in more than 12% of baskets. Support then declines rapidly, and 44 of the 119 products appear in fewer than 1% of transactions. These staple products strongly influence any similarity measure between baskets, which becomes visible in the clustering results.

1.4.3 Principal Component Analysis

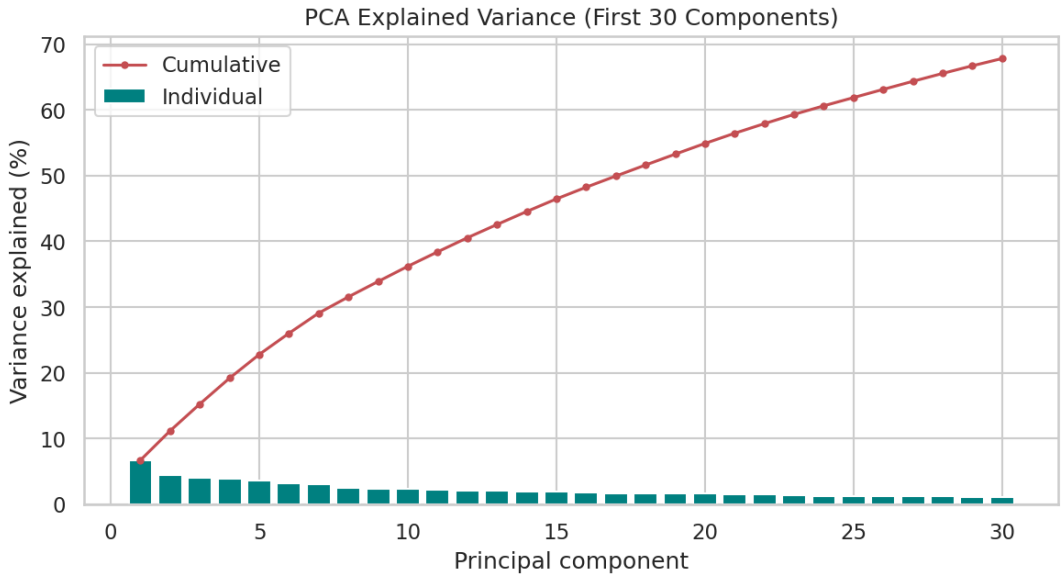


Figure 3. Variance explained by the first 30 principal components.

Interpretation: no component dominates. PC1 explains only 6.7% of the variance and PC2 4.5%. Eighteen components are required to reach 50% of the variance, 33 for 70% and 66 for 90%. Baskets therefore do not fall along a few strong shopping "themes"; purchases of different products are largely independent. The leading components are driven by the most popular products. PC1 loads most heavily on mineral water, spaghetti, milk, chocolate and ground beef, and PC2 on french fries, eggs, mineral water, chocolate and green tea. The 10 components retained for clustering keep **36.2%** of the total variance.

1.5 Model Evaluation

1.5.1 Selecting the number of clusters for K-means

k	Inertia	Silhouette	Davies–Bouldin
2	8230.6	0.254	1.947
3	7371.8	0.251	1.791
4	6651.1	0.263	1.670
5	6122.9	0.268	1.560
6	5728.0	0.277	1.520
7	5375.2	0.291	1.477
8	5071.6	0.307	1.561
9	4820.9	0.311	1.535
10	4679.9	0.322	1.563

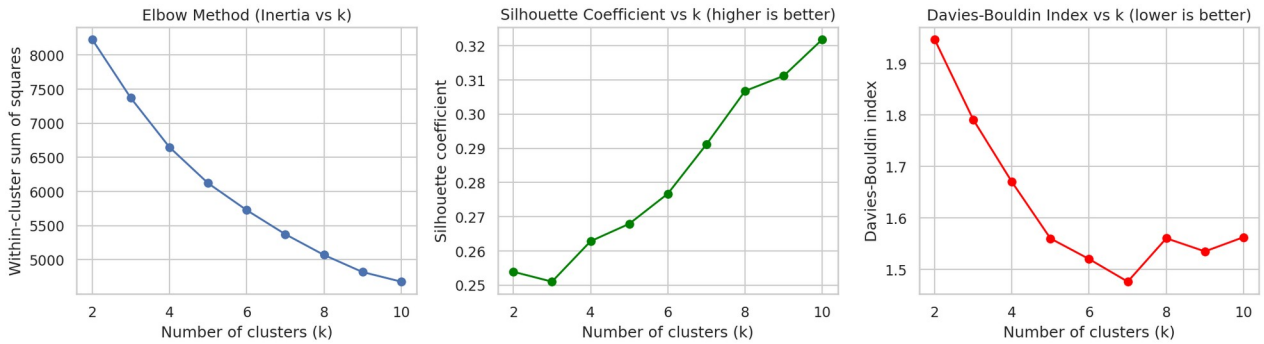


Figure 4. Elbow curve, silhouette coefficient and Davies–Bouldin index against k.

Interpretation: inertia decreases smoothly without a sharp elbow, so the elbow method alone is inconclusive. The silhouette coefficient increases slowly with k, from 0.25 to 0.32, but it never reaches a clear peak. The Davies–Bouldin index reaches its minimum of **1.477 at k = 7**. k = 7 was therefore chosen: it produces the most compact and well-separated clusters according to DB, has a reasonable silhouette (0.291) and gives a number of groups a business can interpret.

1.5.2 Interpreting the K-means clusters

Cluster	Baskets	Avg. items	Defining products (% of baskets in cluster)
0	872	4.89	Chocolate (100%), mineral water (32%), french fries (23%)
1	732	4.03	Eggs (100%), mineral water (27%), burgers (14%)
2	2,546	2.12	Mixed: cookies (13%), escalope (9%), milk (8%)
3	566	4.54	Green tea (100%), chocolate (16%), eggs (16%)
4	858	4.45	Mineral water (100%), milk (15%), french fries (13%)
5	1,113	6.45	Spaghetti (100%), mineral water (40%), ground beef (26%)
6	814	3.91	French fries (100%), eggs (22%), green tea (17%)

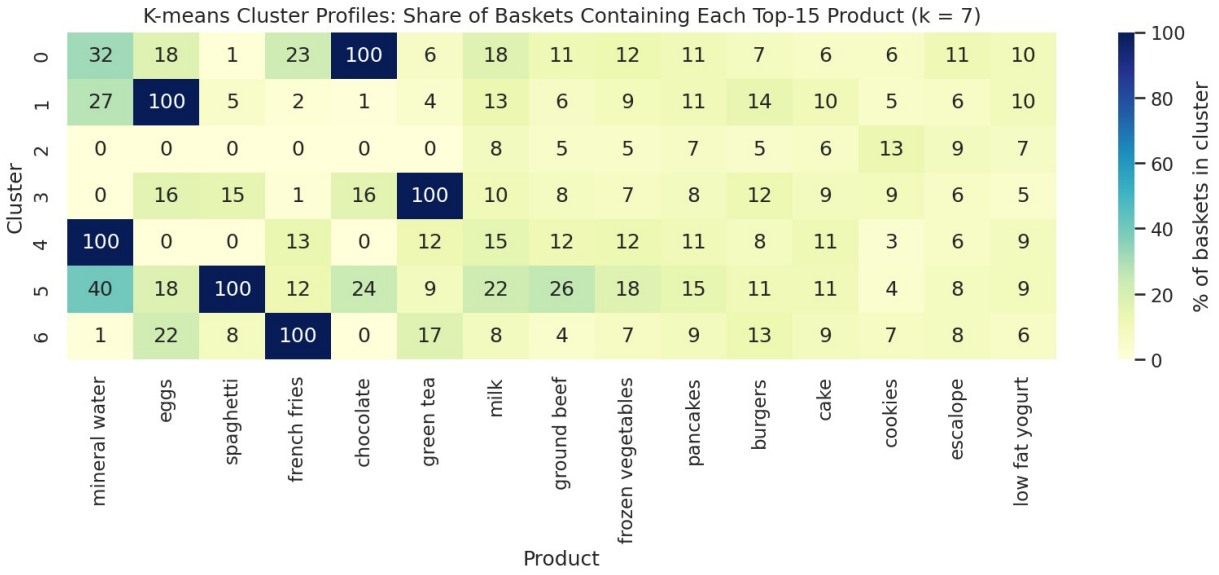


Figure 5. Share of baskets in each K-means cluster that contain each top-15 product.

Interpretation: six clusters are each "anchored" by one staple product that appears in 100% of their baskets: chocolate, eggs, green tea, mineral water, spaghetti and french fries. The spaghetti cluster has the largest baskets (6.45 items on average) and frequently also contains mineral water and ground beef, which resembles a meal-preparation shop. The largest cluster (2,546 baskets, 2.12 items on average) contains none of the anchor products. It is a mixed group of small baskets of less common items.

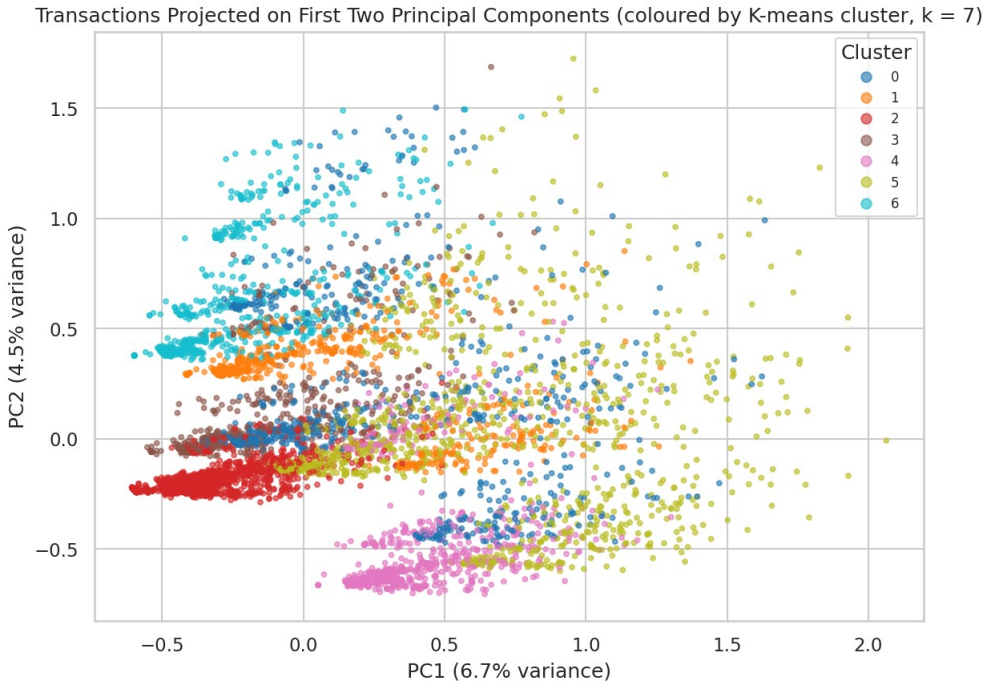


Figure 6. Transactions on the first two principal components, coloured by K-means cluster.

Interpretation: several clusters appear as distinct bands in the PC1–PC2 plane. For example, the french-fries cluster lies at high PC2 values and the eggs cluster forms its own horizontal band, while the mixed small

baskets sit at the low-PC1 end. Some overlap is expected because these two components show only 11.2% of the variance. The clusters are better separated in the 10-dimensional space used for clustering.

1.5.3 Hierarchical, DBSCAN and spectral clustering

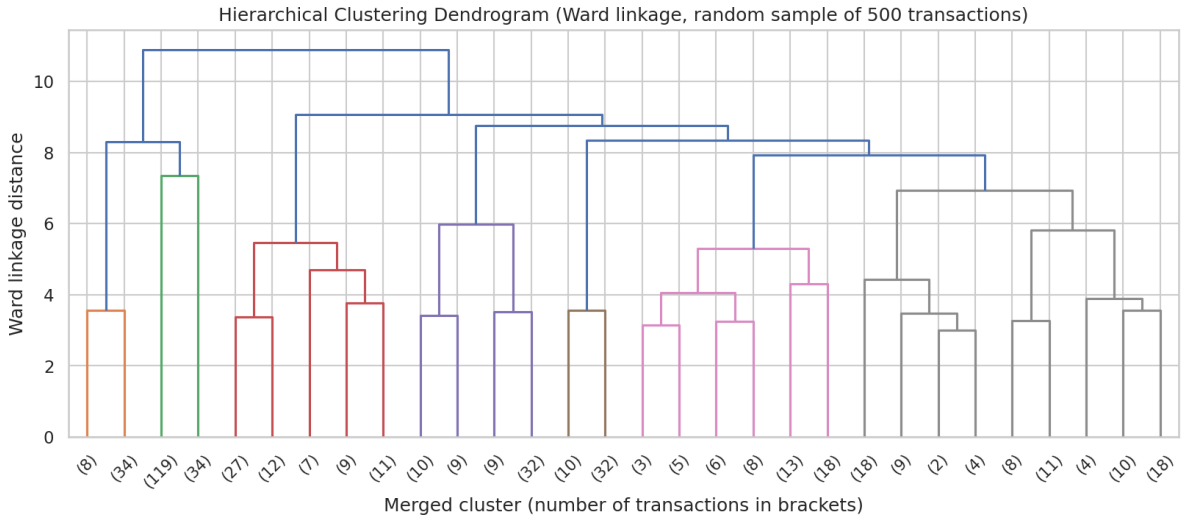


Figure 7. Ward-linkage dendrogram for a random sample of 500 transactions.

Hierarchical clustering: the dendrogram shows several large merges at high linkage distances, consistent with a few broad groups. With k = 7, Ward's method produced two large clusters of about 1,800–2,000 baskets. It splits the mixed small baskets differently from K-means, but its partition still agrees moderately with K-means (ARI = 0.504).

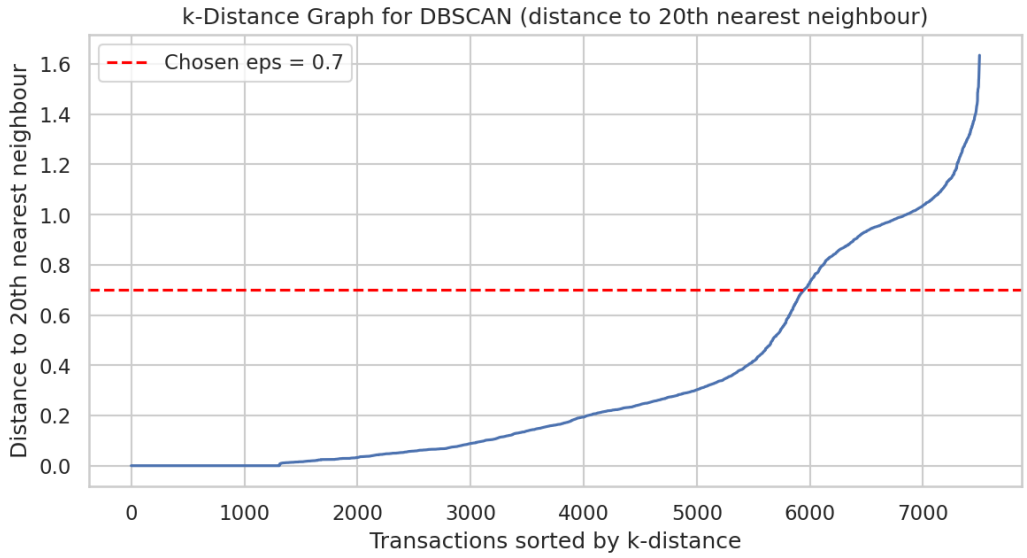


Figure 8. k-distance graph used to choose eps for DBSCAN (min_samples = 20).

DBSCAN: the k-distance curve stays low for about 80% of transactions and then bends upwards at around 0.7, so eps = 0.7 was chosen. A sensitivity check showed that eps = 0.5 gives 36 clusters with 1,628 noise points, while eps = 0.9 merges almost everything into a single cluster. With eps = 0.7, DBSCAN found **33 clusters** and labelled **1,118 baskets (14.9%) as noise**. Many of its clusters are tight groups of near-identical small baskets,

which reflects the many exact repeats in the data. DBSCAN therefore finds dense pockets of identical behaviour rather than broad shopping segments.

Spectral clustering: this produced one very large cluster of 4,144 baskets and six small ones. The nearest-neighbour graph is dominated by the dense centre of small baskets, so the method does not split it effectively. Its agreement with K-means is low (ARI = 0.148).

1.5.4 Comparison of clustering algorithms

Algorithm	Clusters	Baskets scored	Silhouette ↑	Davies–Bouldin ↓
K-means	7	7,501	0.291	1.477
Hierarchical (Ward)	7	7,501	0.212	1.988
DBSCAN (noise excluded)	33	6,383	0.528	0.787
Spectral	7	7,501	0.058	1.964

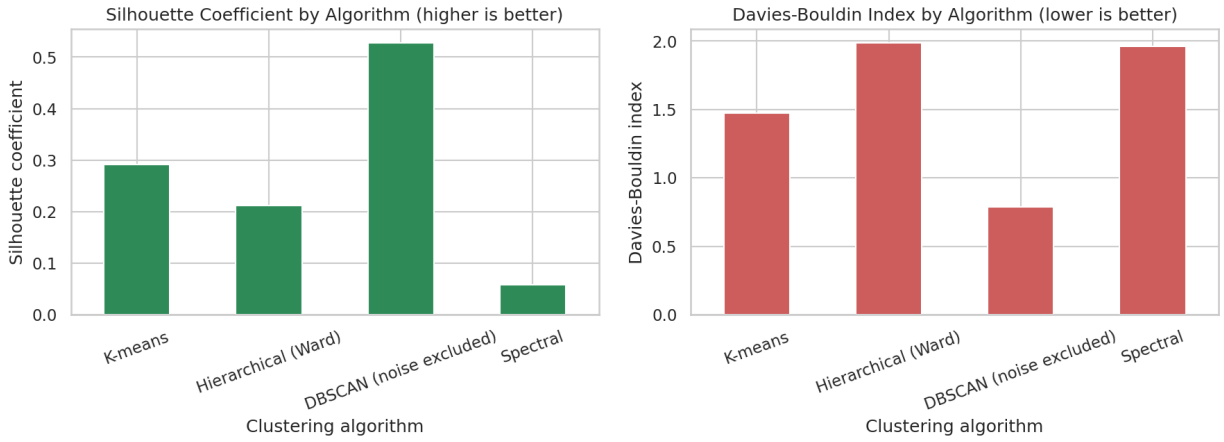


Figure 9. Silhouette coefficient and Davies–Bouldin index by algorithm.

Interpretation:

- **K-means performs best among the algorithms that assign every basket**, with the highest silhouette (0.291) and the lowest DB index (1.477).
- **Hierarchical clustering** is weaker on both measures (0.212 and 1.988). **Spectral clustering** is the weakest: its silhouette close to zero (0.058) means its clusters overlap heavily.
- **DBSCAN's apparently superior scores are not a like-for-like comparison.** They are calculated only on the 85% of baskets that are not noise, and its 33 clusters are largely groups of identical baskets, which are trivially compact.
- All silhouette values are well below 0.5. Even the best partition therefore represents **weak natural structure**: the baskets overlap substantially, and the clusters are useful summaries rather than sharply separated customer types.

1.6 Limitations and Business Implications

Limitations:

1. Binary encoding ignores quantity, price and purchase time; two baskets are considered similar only if they share products.
2. Only 36.2% of the variance is retained in 10 PCA components, so some information about rarer products is lost.
3. Low silhouette scores (below 0.33 for all methods that assign every basket) indicate weak cluster structure.
4. Clusters describe transactions, not customers. There is no customer identifier, so repeat buying behaviour cannot be studied.
5. Results depend on parameter choices (k, eps, neighbourhood size) and on using Euclidean distance on PCA scores. Jaccard distance is a common alternative for basket data.

Business implications:

- The six anchor products (mineral water, eggs, spaghetti, french fries, chocolate and green tea) appear to drive store visits. They are natural candidates for placement next to complementary products or for featured promotions.
- Spaghetti baskets are the largest and often include ground beef and mineral water, which supports "meal bundle" offers.
- Nearly a quarter of transactions contain a single item. Add-on recommendations at checkout could increase basket size.
- Association-rule mining (e.g. Apriori) is a natural next step to quantify which specific product pairs are bought together.

1.7 Conclusion

The retail data consist of many small baskets dominated by a few staple products. PCA showed that the variance is spread across many dimensions, so transactions do not separate into a few strong themes. Among the four algorithms, K-means with k = 7 gave the most useful and best-scoring segmentation (silhouette 0.291, Davies–Bouldin 1.477). It identified six staple-anchored segments and one mixed segment of small baskets. Hierarchical clustering partly confirmed this structure, spectral clustering failed to separate the dense centre, and DBSCAN found only micro-groups of identical baskets. Because the overall cluster structure is weak, the segments are best used as practical summaries for merchandising decisions and as a starting point for association-rule analysis.

Part II — Case Study 4: Student Dropout and Academic Success

2.1 Problem Statement and Objectives

A higher education institution wants to understand the factors associated with student dropout and academic success. It has collected data on students' demographic characteristics, previous educational qualifications, admission information, socioeconomic circumstances and academic performance. The institution wants to use data science techniques to analyse these records and develop a model that predicts whether a student's eventual academic outcome is **Dropout**, **Enrolled** or **Graduate**, so that it can investigate patterns of academic difficulty and plan student-support initiatives. Because the target records the status at the end of the normal course duration, only variables that would genuinely be available at the time of an early prediction should be used.

Objectives:

- Load, understand and preprocess the data, and categorise attributes by their meaning.
- Explore outcome patterns statistically and visually, focusing on admission grade, first-semester performance and age.
- Test formally whether first-semester grades differ across outcomes, distinguishing statistical significance from practical importance.
- Build and evaluate a multiclass logistic regression model using macro-averaged metrics, considering the effect of class imbalance.
- Interpret the model, state its limitations and recommend responsible, evidence-based interventions.

2.2 Dataset Description and Preprocessing

The dataset (Studentdata.csv, from the UCI Machine Learning Repository) contains **4,424 student records and 37 columns**: 36 predictors and the target. The file is **semicolon-separated**, so it was read with sep = ";". One column name contained a hidden tab character, which was removed, and the original spelling "Nacionality" was corrected to "Nationality".

2.2.1 Target and attribute types

The target variable is **Target**, with three classes. Attributes were categorised by what their encoded values *mean*, not by their stored integer type:

Type	Attributes	Treatment
Categorical (nominal codes)	Marital status, application mode, course, previous qualification, nationality, mother's and father's qualification and occupation (9)	One-hot encoding
Binary (0/1)	Daytime/evening attendance, displaced, educational special needs, debtor, tuition fees up to date, gender, scholarship holder, international (8)	Used as 0/1 and scaled
Numerical	Application order, previous qualification grade, admission grade, age at enrollment, 1st- and 2nd-semester curricular unit counts and grades, unemployment rate, inflation rate, GDP (19)	Standard scaling

For example, Course = 9254 is a label for a degree programme, not a quantity, so treating it as a number would wrongly imply that one course is "greater" than another.

2.2.2 Data quality checks

- There are **no missing values and no duplicate records**.
- **718 students have a first-semester grade of 0.** All 718 approved zero units, and 180 of them were not enrolled in any units. Grades are on a 0–20 scale with 10 as the pass mark, so a 0 means "no grade obtained" rather than an actual mark. These records were kept because they are highly informative, and the hypothesis test was repeated without them as a sensitivity check.
- Age at enrollment ranges from 17 to 70. 441 students are IQR outliers and 225 are over 40. These are plausible mature students, not data errors, so they were kept.
- Admission grades lie within their valid range of 95 to 190.

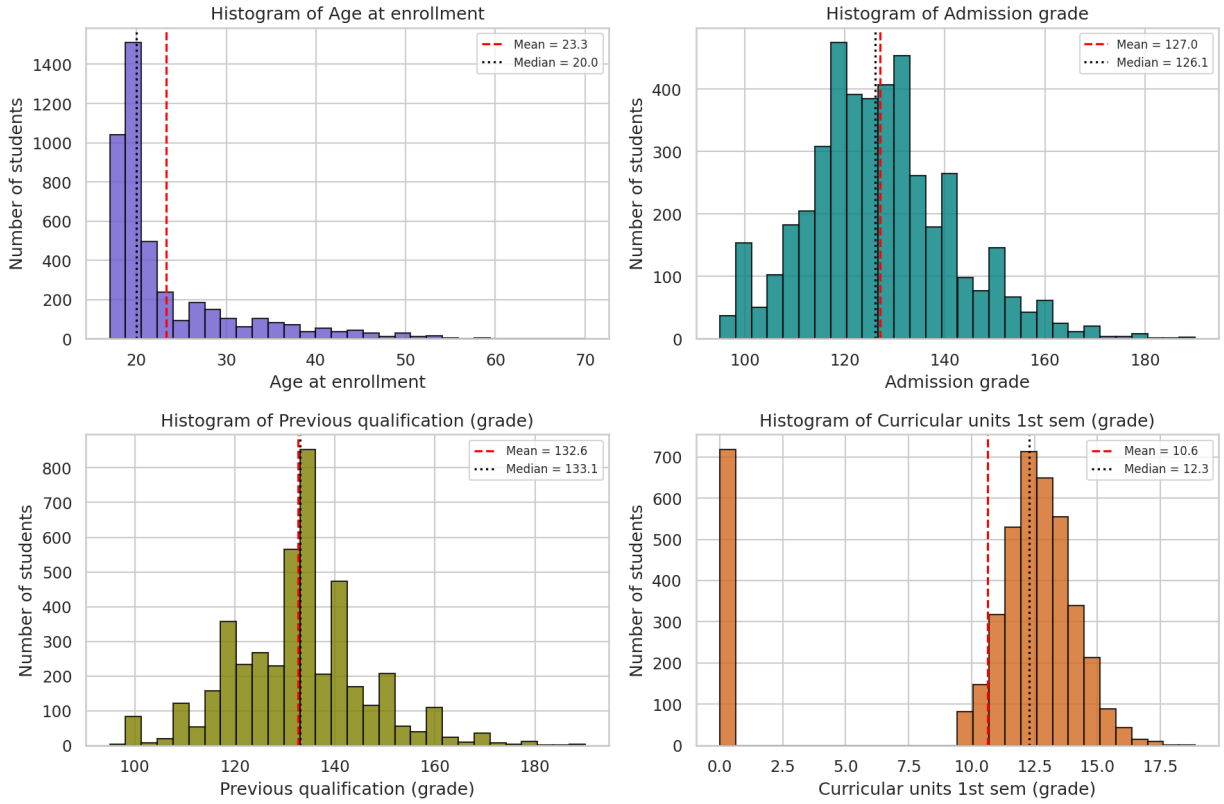


Figure 10. Histograms of age at enrollment, admission grade, previous qualification grade and first-semester grade.

Interpretation: age at enrollment is strongly right-skewed (skewness 2.05). Most students begin at 18–20, with a long tail of mature students. Admission grade and previous-qualification grade are roughly bell-shaped, centred around 125–135. The first-semester grade is left-skewed (skewness −1.57) and bimodal: most students score between 11 and 14, with a separate spike at 0 representing students with no grade.

2.2.3 Preprocessing decisions

1. **One-hot encoding** of the nine categorical code columns; categories with fewer than 10 students were grouped to avoid unstable estimates.
2. **Standard scaling** of numerical and binary features, because regularised logistic regression is sensitive to feature scale and scaling makes coefficients comparable.

3. **Exclusion of all six 2nd-semester variables** for an early-warning model, since they are not available at the end of semester 1. Including them would leak future information and inflate performance.
4. **Stratified train/test split** so that class proportions are identical in both sets.
5. **Fitting all preprocessing inside a Scikit-learn Pipeline** on the training data only, preventing test-set information from leaking into the encoder or scaler.

2.3 Methodology

The analysis proceeded in five stages: (i) data understanding and preprocessing, (ii) exploratory data analysis with descriptive statistics, visualisations, a Kruskal–Wallis test for admission grade and a chi-square test for age group, (iii) formal hypothesis testing of first-semester grades with assumption checks, a post-hoc test and a sensitivity analysis, (iv) training a multinomial logistic regression with Scikit-learn and evaluating it with a confusion matrix and per-class and macro-averaged metrics, and (v) interpretation of coefficients and recommendations. The core model is defined as follows:

```
pre = ColumnTransformer([('cat', OneHotEncoder(handle_unknown='ignore',
min_frequency=10), cat_cols),
                         ('num', StandardScaler(), num_cols)])
model = Pipeline([('prep', pre), ('clf', LogisticRegression(max_iter=3000))])
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, stratify=y,
random_state=42)
model.fit(X_train, y_train);  y_pred = model.predict(X_test)
```

Purpose: the ColumnTransformer encodes categorical columns and scales numerical ones. The Pipeline guarantees that these transformations are learned from the training set only. With three classes, LogisticRegression fits a multinomial (softmax) model that outputs a probability for each outcome. *Observation:* after encoding, the model uses 126 input features from 30 original variables.

Train/test strategy: an 80/20 split gives 3,539 training and 885 test students. Stratification keeps the class proportions identical in both sets (Dropout 32.1%, Enrolled 18.0%, Graduate 49.9%). This matters because, without it, the small Enrolled class could be under-represented in the test set and its metrics would be unreliable.

2.4 Statistical Findings

2.4.1 Distribution of outcomes

Outcome	Students	Percentage
Dropout	1,421	32.12%
Enrolled	794	17.95%
Graduate	2,209	49.93%
Total	4,424	100%

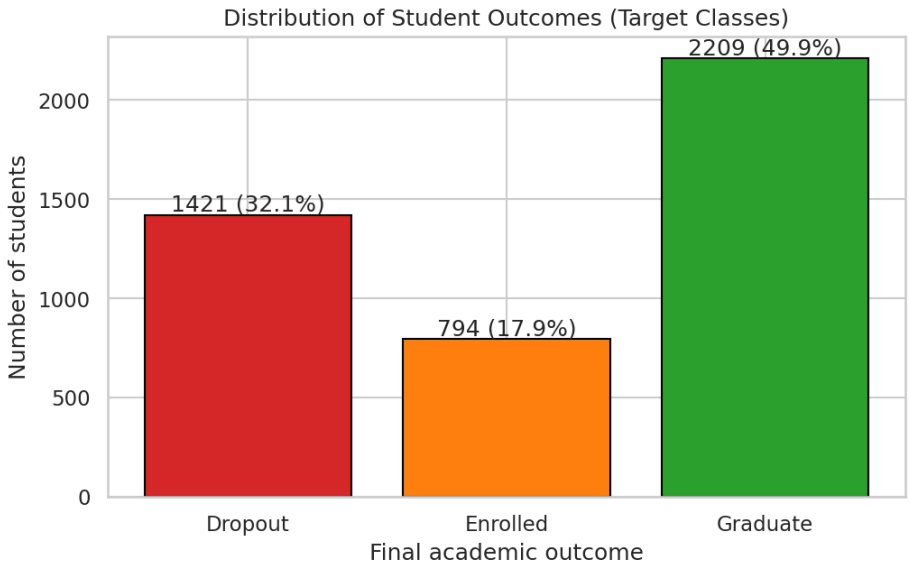


Figure 11. Distribution of student outcomes.

Interpretation: about half the students graduate, almost one third drop out and 18% are still enrolled. The classes are **imbalanced**: the Enrolled class is less than half the size of the Graduate class. A model that always predicted "Graduate" would already be about 50% accurate, so accuracy alone cannot be used to judge the model.

2.4.2 Descriptive statistics by outcome

Variable	Overall mean	Dropout	Enrolled	Graduate
Age at enrollment	23.27	26.07	22.37	21.78
Admission grade	126.98	124.96	125.53	128.79
Previous qualification grade	132.61	131.11	131.21	134.08
1st-sem grade	10.64	7.26	11.13	12.64
1st-sem units approved	4.71	2.55	4.32	6.23
1st-sem units enrolled	6.27	5.82	5.96	6.67

Dropouts are on average four years older than graduates and approve far fewer first-semester units (2.55 compared with 6.23). Their mean first-semester grade (7.26) is more than five points below that of graduates. Differences in admission grade are comparatively small.

2.4.3 Admission grade across outcomes

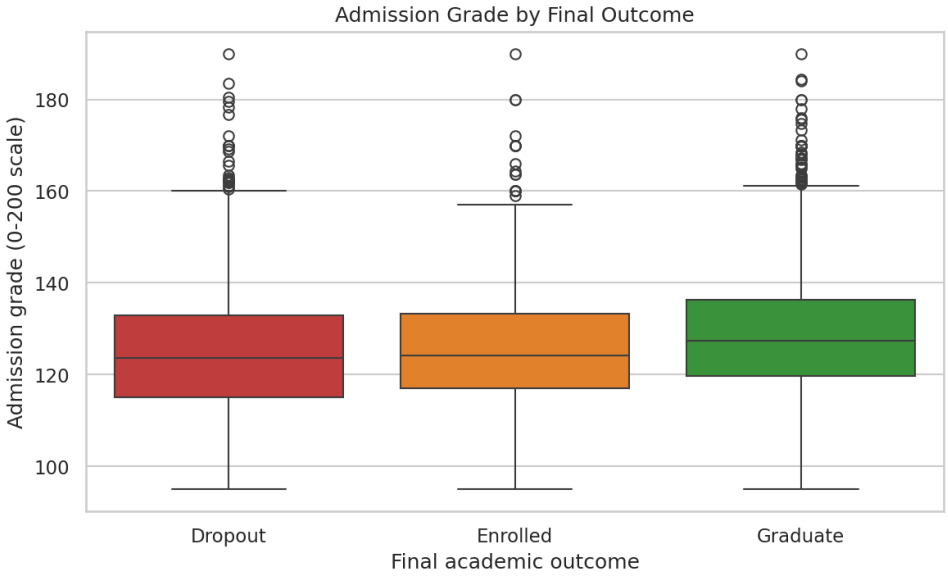


Figure 12. Admission grade by final outcome.

Interpretation: graduates have a slightly higher median admission grade (127.4) than dropouts (123.6) and enrolled students (124.1), but the three boxes overlap almost completely. A Kruskal–Wallis test shows that the difference is statistically significant ($H = 73.23$, $p \approx 1.3 \times 10^{-16}$). However, the effect size is only $\varepsilon^2 = 0.016$: outcome accounts for under 2% of the variation in admission-grade ranks. **The difference is real but of little practical importance**, so admission grade alone is a weak indicator of who will drop out.

2.4.4 First-semester performance and final outcome

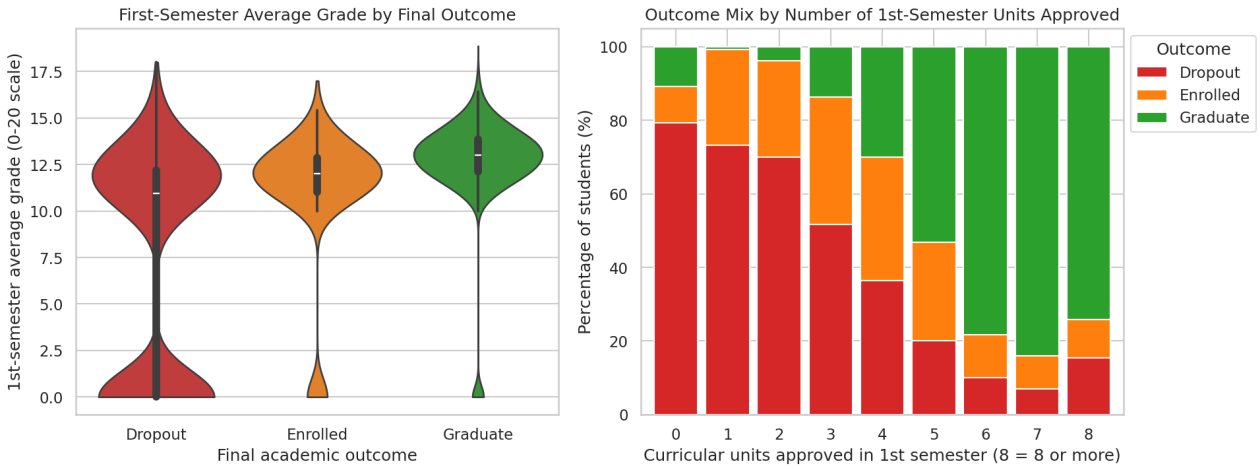


Figure 13. First-semester grade by outcome (left) and outcome mix by units approved in semester 1 (right).

Interpretation: graduates' first-semester grades are concentrated between 12 and 14. Dropouts form two groups: a large group with a grade of 0 (40.1% of dropouts) and another around 11–12. The stacked bar chart shows the dropout share falling steadily as more units are approved: **79.4% of students who approved no first-semester units dropped out, compared with 10.2% of those who approved six or more.** First-semester academic progress is the clearest early signal of final outcome in this dataset.

2.4.5 Age at enrollment and outcome

Age group	Students	Dropout %	Enrolled %	Graduate %
17–20	2,551	21.2	18.5	60.3
21–24	735	33.7	21.9	44.4
25–29	427	57.6	16.9	25.5
30–39	463	55.7	13.0	31.3
40+	248	51.2	12.1	36.7

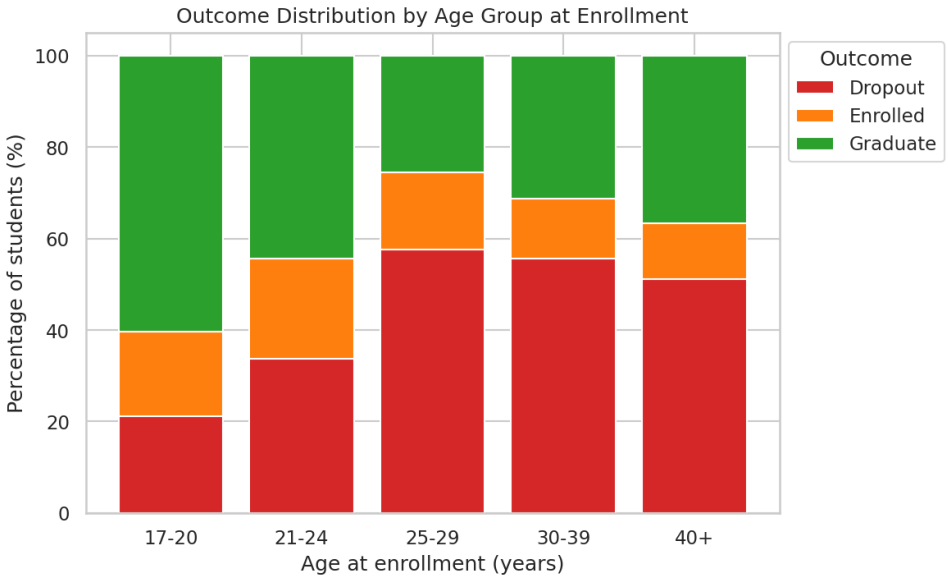


Figure 14. Outcome distribution by age group at enrollment.

Interpretation: the outcome mix changes clearly with age. 60.3% of students who enrolled at 17–20 graduated, while more than half of those who enrolled at 25 or older dropped out. A chi-square test of independence confirms a significant association of moderate strength ($\chi^2 = 458.71$, df = 8, $p < 0.001$, Cramér's V = 0.228). This is an association only. Mature students are more likely to be studying while working or caring for family, and such circumstances, rather than age itself, may explain the pattern.

2.4.6 Tuition-fee status

86.6% of students whose tuition fees were not up to date dropped out, compared with 24.7% of students whose fees were up to date. Financial difficulty is therefore strongly associated with dropout. Because the dataset does not document when this variable was recorded, it may partly reflect a decision to leave rather than precede it.

2.5 Hypothesis Testing

Research question: do first-semester average grades differ between students who drop out, remain enrolled and graduate?

- **Null hypothesis (H_0):** the distribution of first-semester grades is the same for Dropout, Enrolled and Graduate students.

- **Alternative hypothesis (H_1):** the distribution differs for at least one outcome group.
- **Significance level:** $\alpha = 0.05$.

2.5.1 Selecting the test and checking assumptions

One-way ANOVA is the standard test for comparing three group means. It assumes that the data in each group are approximately normal and that the group variances are equal. Both assumptions were checked:

Group	n	Std. dev.	Skewness	Normality test p-value
Dropout	1,421	6.03	−0.31	< 0.001
Enrolled	794	3.68	−2.36	< 0.001
Graduate	2,209	2.70	−3.45	< 0.001

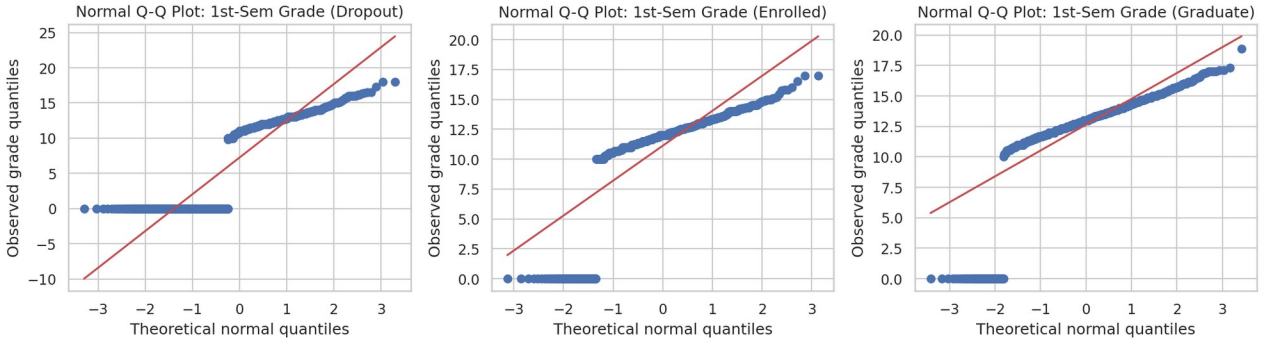


Figure 15. Normal Q-Q plots of first-semester grade for each outcome group.

Interpretation: in all three Q-Q plots the points bend sharply away from the reference line, mainly because of the spike at grade 0, and D'Agostino's normality test rejects normality for every group ($p < 0.001$). Levene's test shows strongly unequal variances ($W = 549.87$, $p < 0.001$); the dropout group's standard deviation is more than twice that of graduates. **The assumptions of ANOVA are not reasonable**, so the non-parametric **Kruskal–Wallis H test** was selected. It compares the rank distributions of the groups and requires only independent observations, an ordinal or continuous variable and independent groups. All of these hold, since each student appears once and belongs to exactly one outcome.

2.5.2 Test results

Quantity	Value
Test statistic H	1094.19
Degrees of freedom	2
p-value	2.5×10^{-238}
Effect size ε^2	0.247 (large)
Decision at $\alpha = 0.05$	Reject H_0

Conclusion: since $p < 0.05$, the null hypothesis is rejected. There is very strong evidence that first-semester grades differ between students who drop out, remain enrolled and graduate.

Statistical significance versus practical importance: with 4,424 students, even very small differences can be statistically significant, as the admission-grade result in Section 2.4.3 showed. The effect size is therefore essential. Here $\varepsilon^2 = 0.247$ means that about a quarter of the variation in grade ranks is associated with outcome, which is a large effect. This difference is both statistically significant **and** practically important. One-way ANOVA, shown only for reference, reaches the same conclusion ($F = 713.52$, $p < 0.001$).

2.5.3 Post-hoc comparisons and sensitivity analysis

Comparison	Median difference	Adjusted p-value	Rank-biserial r
Dropout vs Enrolled	1.07	< 0.001	0.35
Dropout vs Graduate	2.07	< 0.001	0.62
Enrolled vs Graduate	1.00	< 0.001	0.40

All three pairs of groups differ significantly. The largest separation is between Dropout and Graduate students (rank-biserial r = 0.62, a large effect), with enrolled students in between. When the 718 students with a grade of 0 are excluded, the test remains highly significant ($p \approx 10^{-104}$), but the effect size halves to $\varepsilon^2 = 0.128$ and the medians become close (12.0, 12.1 and 13.0). Much of the original effect therefore comes from students who obtained **no** grade. Not engaging with assessments is an even stronger warning sign than receiving a low grade.

2.6 Model Evaluation

The final multinomial logistic regression was trained on 30 early-available variables (all variables except the six 2nd-semester ones) and evaluated on the 885 held-out test students.

2.6.1 Confusion matrix

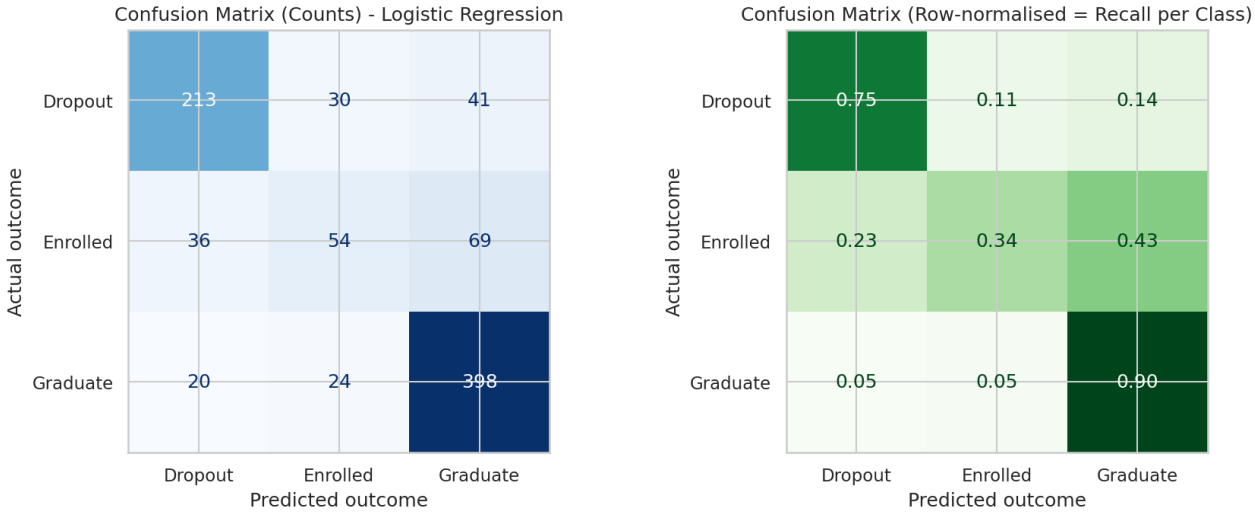


Figure 16. Confusion matrix of the logistic regression model (counts and row-normalised).

Interpretation: rows show the actual outcome and columns the predicted outcome. Of 284 actual dropouts, 213 (75%) were correctly identified; 30 were predicted as Enrolled and 41 as Graduate. Of 442 graduates, 398 (90%) were correct. **Enrolled was the hardest class**: only 54 of 159 (34%) were correct, with most predicted

as Graduate (69) or Dropout (36). This is understandable, because students still enrolled are mid-way and their records resemble either eventual dropouts or eventual graduates.

2.6.2 Classification performance

Class	Precision	Recall	F1-score	Support
Dropout	0.792	0.750	0.770	284
Enrolled	0.500	0.340	0.404	159
Graduate	0.783	0.900	0.838	442
Macro average	0.692	0.663	0.671	885
Accuracy			0.751	885

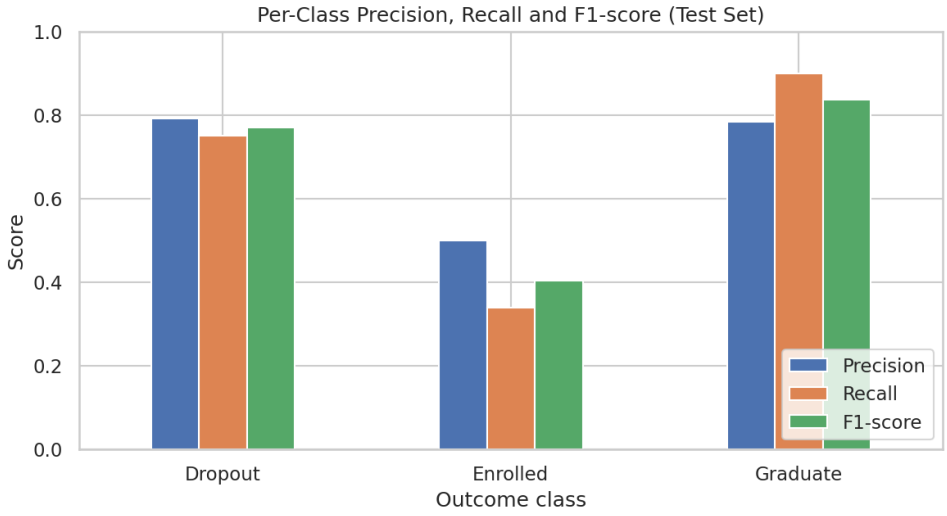


Figure 17. Per-class precision, recall and F1-score on the test set.

Interpretation: *precision* is the proportion of students predicted as a class who truly belong to it. *Recall* is the proportion of a class's students the model identifies. The *F1-score* combines the two. Macro-averaged metrics weight each class equally, so the minority class counts as much as the majority.

- Overall accuracy is 0.751, well above the 0.499 baseline of always predicting "Graduate". The macro F1-score is 0.671.
- **Graduate** (F1 0.838) and **Dropout** (F1 0.770) students are predicted well. When the model flags a student as a likely dropout, it is correct 79% of the time.
- **Enrolled** students are predicted poorly (F1 0.404, recall 0.340): two-thirds of them are missed.

The gap between accuracy (0.751) and macro F1 (0.671) comes from this weak minority class. The model performs very differently across the three classes, which the single accuracy figure hides.

2.6.3 Effect of class imbalance

Model	Accuracy	Macro F1	Recall Dropout	Recall Enrolled	Recall Graduate
Unweighted (final)	0.751	0.671	0.750	0.340	0.900
Class-weighted (balanced)	0.690	0.652	0.669	0.566	0.749
Admission-time features only	0.629	0.521	–	–	–

Interpretation: to test whether imbalance was harming the minority class, a second model was trained with class_weight = "balanced", which gives more weight to errors on the smaller classes. This raised Enrolled recall from 0.340 to 0.566, but it reduced Dropout recall (0.750 → 0.669) and Graduate recall (0.900 → 0.749), and both accuracy and macro F1 fell. The unweighted model was kept as the final model because it has the higher macro F1 and identifies more of the students who actually drop out, which is the main purpose of an early-warning system.

This comparison shows why accuracy alone is misleading on imbalanced data: an accuracy of 0.751 conceals an Enrolled recall of only 34%. Any re-weighting is a trade-off between classes and should follow the institution's priorities. Finally, a model using only admission-time information reached a macro F1 of just 0.521, about 0.15 lower. First-semester results provide the decisive information, so the end of semester 1 is the most useful point for an early check.

2.7 Interpretation, Limitations and Recommendations

2.7.1 Model coefficients

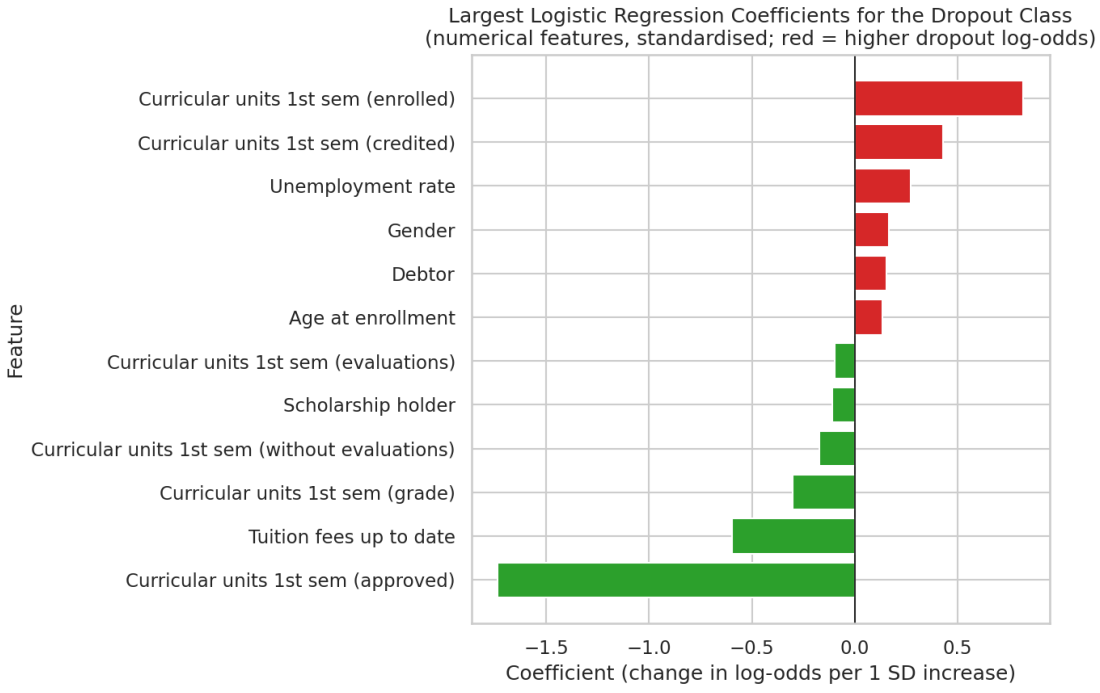


Figure 18. Largest standardised logistic regression coefficients for the Dropout class.

Because the numerical features were standardised, each coefficient gives the change in the log-odds of dropout for a one-standard-deviation increase in that feature, holding the others fixed. The exponential of a coefficient is the corresponding odds multiplier.

Feature	Coefficient (Dropout)	Odds multiplier per SD
1st-sem units approved	−1.735	0.18
Tuition fees up to date	−0.598	0.55
1st-sem grade	−0.300	0.74
Scholarship holder	−0.110	0.90
Debtor	+0.152	1.16
Age at enrollment	+0.134	1.14
Unemployment rate	+0.273	1.31
1st-sem units credited	+0.428	1.53
1st-sem units enrolled	+0.820	2.27

- **Units approved in semester 1** has by far the strongest association: each additional standard deviation of approved units multiplies the odds of dropout by 0.18.
- **Units enrolled** has a positive coefficient, but it must be read together with units approved: holding approvals fixed, enrolling in more units means failing more of them. The underlying signal is the **pass rate**.
- **Fees being up to date** and **holding a scholarship** are associated with lower dropout odds, while **debtor status**, **older age at enrollment** and a **higher unemployment rate** are associated with higher odds.

These coefficients describe associations within the model, and correlated first-semester variables make individual values somewhat unstable. **They do not establish causation.** For example, the model does not show that age or financial status causes dropout, only that these characteristics co-occur with it in this data.

2.7.2 Limitations

1. **Poor performance on the Enrolled class** (F1 0.40), and about 25% of actual dropouts are still missed.
2. **Linear model:** logistic regression assumes additive effects on the log-odds and cannot capture interactions (for example, between age and course) without manual feature engineering.
3. **Correlated predictors:** the first-semester unit counts overlap, making individual coefficients unstable and harder to interpret.
4. **Possible timing leakage:** the dataset does not document when "tuition fees up to date" and "debtor" were recorded. If they were recorded late, they may partly reflect decisions already taken.
5. **Limited generalisability:** the data come from one institution and the model was evaluated on a single 80/20 split. Cross-validation and external validation would be needed before wider use.
6. **"Enrolled" is not a final state:** some currently enrolled students will later graduate or drop out, so this class label is inherently noisy.

2.7.3 Evidence-based recommendations

1. **End-of-semester-1 academic review:** contact every student who approved zero to two first-semester units, the group in which dropout rates are highest (79.4% when no units are approved). Offer tutoring, study-skills support and advice on course load.
2. **Financial support pathway:** students whose fees are not up to date drop out at 86.6%. The institution could investigate proactive outreach about payment plans, emergency grants and scholarship eligibility.
3. **Support for mature students:** students enrolling at 25 or older drop out at more than 50%. Flexible timetabling, evening study options and peer mentoring for students balancing work or family could be piloted and evaluated.
4. **Monitor engagement, not only grades:** a grade of 0 (no assessments passed) is a stronger warning sign than a low pass mark, so attendance and assessment-submission data should feed into early alerts.

2.7.4 Responsible use of the model

- Use predictions only to **offer support**, never to penalise, exclude or limit a student's opportunities.
- Keep a **human in the loop**: academic advisers should review each flag together with context the data cannot capture.
- **Audit fairness:** check whether error rates differ across groups defined by gender, nationality, age or parental background, and consider excluding sensitive attributes that do not improve support decisions.
- Be **transparent** with students about which data are used, and allow them to correct their records.
- **Re-validate regularly**, because student cohorts and economic conditions change over time.

2.8 Conclusion

First-semester academic progress is the strongest early indicator of whether a student will drop out, remain enrolled or graduate. The Kruskal–Wallis test confirmed a statistically significant and practically large difference in first-semester grades across outcomes ($\varepsilon^2 = 0.247$). In contrast, admission grades differ significantly but only trivially. Age and tuition-fee status are also clearly associated with outcome. A multinomial logistic regression using only information available at the end of semester 1 achieved 75.1% accuracy and a macro F1-score of 0.671, predicting dropouts and graduates well but enrolled students poorly. Per-class and macro-averaged evaluation was therefore essential. Used transparently, with human review and fairness checks, such a model can help the institution direct academic, financial and mature-student support to students who may benefit most, without unfairly labelling any student.

Implementation and Code Availability

Both case studies were implemented in a single Jupyter Notebook, **Case_Studies_1_and_4.ipynb**. **Part A** contains Case Study 1 and **Part B** contains Case Study 4. Each code cell is preceded by an explanation of its purpose and followed by the observations obtained from its output. The same code is provided as a standalone Python script, **case_studies_1_and_4.py**. All results in this report were produced by executing the notebook with a fixed random seed (random_state = 42), so they can be reproduced exactly.

Libraries used: Pandas, NumPy, Matplotlib, Seaborn, SciPy and Scikit-learn.

Data files: Online_retail_new.csv (Case Study 1) and Studentdata.csv (Case Study 4), placed in the same folder as the notebook.

GitHub repository: _____

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